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Chapter 4

Planning the Measurements: Precision, Accuracy, and Validity

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Measurements describe phenomena in terms that can be analyzed statistically, and the validity of a study depends on how well the variables designed for—and measured in—the study represent the phenomena of interest (Figure 4.1), for example, how well does parental recall of birth weight measure actual birth weight? (1) These concepts are important for all types of measurements, including physical examinations, laboratory tests, and scores from questionnaires.

This chapter begins by considering how the choice of measurement **scale** influences the information content of the measurement. We then turn to the central goal of maximizing test **accuracy** by designing measurements that are relatively **precise** (free of random error) and **unbiased** (free of systematic error), thereby enhancing the appropriateness of drawing inferences from these measurements to the phenomena of interest. We address the concept of **validity**, a qualitative relative of accuracy, before concluding with some considerations for clinical and translational research, noting especially the advantages of storing specimens for later measurements.

The topics discussed in this chapter apply to the study's predictor and outcome variables, as well as to the study's **covariates**, the remaining set of measured variables. The measurement methods for some covariates will not require detailed planning, because many can be borrowed from previous research. But the investigator will almost always need to give special attention to the study's main predictor(s) and outcome(s)—and possibly a few others—tailoring their measurements to the specific needs of the study.

▪ **FIGURE 4.1** This chapter focuses on the items within the red box: designing measurements that represent the phenomena of interest for the research question.

▪ MEASUREMENT SCALES

Table 4.1 presents a simplified classification of measurement scales and the information they contain. The classification is important because some types of variables are more informative than others, adding power or reducing sample size requirements, and revealing more detailed distribution patterns.

Categorical Variables: Dichotomous, Nominal, and Ordinal

Phenomena that are not suitable for quantification are measured by classifying them in categories. **Categorical variables** with two possible values (eg, dead or alive) are termed **dichotomous**. Categorical variables with more than two categories (**polychotomous**) can be further characterized according to the type of information they contain. Among these, **nominal variables** have categories that are not ordered; type A blood, for example, is neither more nor less than type B blood. Nominal variables tend to have an absolute qualitative character that makes them straightforward to measure. In contrast, **ordinal variables** do have an order, such as severe, moderate, and mild pain. The additional information is an advantage over nominal variables, but because ordinal variables do not specify a numerical or uniform difference or ratio of one category to the next, the information content is less than that of numerical variables.

Numerical Variables: Continuous and Discrete

Numerical variables can be quantified with a number that expresses how much or how many. A **continuous variable**, such as the blood hemoglobin level, quantifies how much on a scale that has a theoretically infinite number of values, although the method used to measure it will likely generate only some of those values. In contrast, a **discrete variable** has a countable number of

values usually expressed as a positive or negative integer, such as the change in the number of people living in a household

during a pandemic. Discrete variables that are well distributed over many possible values, say, six or more, can resemble continuous variables in statistical analyses and be equivalent for the purpose of designing measurements.

TABLE 4.1 MEASUREMENT SCALES

TYPE OF MEASUREMENT	CHARACTERISTICS OF VARIABLE	EXAMPLE	DESCRIPTIVE STATISTICS	STATISTICAL POWER
		Categorical		
Dichotomous	Two categories	Vital status (alive or dead)	Counts, proportions	Low
Nominal	Unordered categories	Gender identity; blood type	Same as above	Low
Ordinal	Ordered categories with intervals that are not quantifiable	Degree of pain (none, mild, moderate, severe); social class	In addition to the above: medians	Intermediate
		Numeric		
Continuous	Infinite number of values	Weight; hemoglobin level	In addition to the above: means, standard deviations	High
Discrete or count[a]	Limited number of values (typically integers)	Number of pregnancies; number of sexual partners	Same as ordinal when there are few possible values; same as continuous	High if the variable takes on many possible values, especially if the average

			when there are many	is near the middle

[a] Count variables are discrete variables that can only be whole numbers (0, 1, 2, 3, …).

Choosing a Measurement Scale

A good general rule is to prefer continuous over categorical variables when there is a choice, because the additional information they contain improves statistical efficiency. In a study comparing the antihypertensive effects of several treatments, for example, measuring blood pressure in millimeters of mercury allows the investigator to observe the magnitude of the change in every participant, whereas measuring it as hypertensive versus normotensive limits the assessment. The continuous variable contains more information, and the result is a study with more power and/or a smaller sample size (Chapters 5 and 6).

Continuous variables also allow for more flexibility than categorical variables in fitting the data to the nature of the variable or the shape of the association, especially when the relationship might have a complex pattern. For example, a study of the association of body mass index (BMI) with mortality may be able to detect a U-shaped pattern, with higher mortality in those with low and high levels of BMI compared with intermediate levels (2) and may be able to describe threshold values where mortality begins to change. And a study of predictors of low-birth-weight babies should record actual birth weight rather than above or below the conventional 2500 g threshold for low birth weight; this leaves analytic options open, such as to change the cutoff that defines low birth weight, to develop an ordinal scale with several categories of birth weight (eg, <1500, 1500 to 1999, 2000 to 2499, and ≥2500 g), or to use the continuous measurement.

Similarly, when there is the option of designing the number of response categories in an ordinal scale, as in a question about satisfaction with care, it is often useful to provide a half-dozen categories that range from “strongly agree” to “strongly disagree.” These results can later be collapsed into a dichotomy (agree and disagree), but the reverse would not be possible.

Many characteristics, such as pain or quality of life, are difficult to describe with categories or numbers, but quantifying them is essential to answering important research questions. This is illustrated by the Short Form (SF)-36, a standardized questionnaire for assessing quality of life that produces discrete numerical ratings of eight dimensions, including physical, social, and emotional function and general well-being (3). The process of classification and measurement, if done well, can increase the objectivity of our knowledge, reduce bias, and provide more detailed communication.

▪ PRECISION

The precision of a variable is the degree to which it is reproducible, with nearly the same value each time it is measured. A beam scale can measure body weight with great precision, whereas measuring balance by timing how long a participant keeps her feet together with eyes closed may produce values that vary from one observer or one occasion to another. Precision has a very important influence on the power of a study. The more precise a measurement, the greater the statistical power at a given sample size to test hypotheses about that measurement (Chapter 5).

Precision (also called **reproducibility**, **reliability**, and **consistency**) is a function of random error (chance variability); the greater the error, the less precise the measurement. There are three main sources of random error in making measurements.

- **Observer variability** is due to the observer and includes such things as choice of words in an interview and skill in using a mechanical instrument.

- **Instrument variability** is due to the instrument and includes changing environmental factors (eg, temperature), differences between batches of reagents, and so on.
- **Participant variability** is due to intrinsic variability in characteristics of study participants unrelated to variables under study that may influence the measurement, such as variability in measurements of strength, due to time of day, time since last food intake, or effort.

Assessing Precision

Precision is assessed as the reproducibility of repeated measurements, comparing measurements made either by the same person (within-observer reproducibility) or different people (between-observer reproducibility). It can also be assessed

within or between instruments. The reproducibility of continuous variables is often expressed as either the within-participant **standard deviation** or the coefficient of variation (within-participant standard deviation divided by the **mean**).[1] For categorical variables, percent agreement, the interclass correlation coefficient, and the **kappa** statistic are often used (4, 5, 6).

Strategies for Enhancing Precision

There are five approaches to minimizing random error and increasing the precision of measurements (Table 4.2):

1. **Standardizing the measurement methods.** All study protocols should include detailed instructions for making the measurements. These may include written directions with pictures or a short video on how to prepare the environment and the participant, how to carry out and record the measurement, how to **calibrate** the instrument, and so forth (Appendix 4A). This set of materials, part of the **operations manual**, is essential for most studies. Even when there is only a single observer, specific written guidelines for making each measurement will help ensure that her performance is uniform over the duration of the study and serve as the basis for describing the methods when the results are published.
2. **Training and certifying the observers.** Training will improve the consistency of measurement techniques, especially when several observers are involved. It is often desirable to test mastery of the techniques specified in the operations manual and to certify that observers have achieved the prescribed level of performance (Chapter 18). When multiple observers make measurements for a study (eg, determining whether a physical finding is present or whether a death was caused by breast cancer), it is helpful to measure and take steps to optimize their interrater reliability (eg, using Kappa; see Chapter 13).
3. **Refining the instruments.** Mechanical and electronic instruments can be engineered to reduce variability. Similarly, questionnaires and interviews should go through multiple iterations to increase clarity and avoid potential ambiguities (Chapter 17).
4. **Automating the instruments.** Variations in the way human observers make measurements can be eliminated with automatic devices and self-response questionnaires.
5. **Repetition.** The influence of random error from any source is reduced by repeating the measurement and using the mean of two or more readings. For example, taking the average of two measurements of leg strength will improve the precision of a measurement that is prone to variability in effort from test to test. Precision will be substantially increased by this strategy, the primary limitations being the added cost and practical difficulties of repeating the measurements as well as the possibility of systematic changes in the measurement with

repetition, for example, due to tiring on strength tests or improved test-taking for tests of cognitive ability.

TABLE 4.2 STRATEGIES FOR REDUCING RANDOM ERROR, WITH ILLUSTRATIONS FROM A STUDY OF ANTIHYPERTENSIVE TREATMENT

STRATEGY TO REDUCE RANDOM ERROR	SOURCE OF RANDOM ERROR	EXAMPLE OF RANDOM ERROR	EXAMPLE OF STRATEGY TO PREVENT THE ERROR
1. Standardizing the measurement methods in an operations manual	Observer	Variation in blood pressure (BP) measurement due to variable rate of cuff deflation (often too fast)	Specify that the cuff be deflated at 2 mm Hg/s

		Participant	Variation in BP due to variable length of quiet sitting before measurement	Specify that participant sit in a quiet room for 5 minutes before BP measurement
2.	Training and certifying the observer	Observer	Variation in BP due to variable observer technique	Train observer in standard techniques
3.	Refining the instrument	Instrument and observer	Variation in BP due to noisy manometer	Purchase new high-quality manometer and periodically test accuracy
4.	Automating the instrument	Observer	Variation in BP due to variable observer technique	Use automatic BP measuring device
		Participant	Variation in BP due to participant's emotional reaction to observer	Use automatic BP measuring device
5.	Repeating the measurement	Observer, participant, and instrument	All measurements and all sources of variation	Use mean of two or more BP measurements

For each measurement in the study, the investigator must decide how vigorously to pursue each of these strategies. This decision can be based on the importance of the variable, the magnitude of the potential problem with precision, and the feasibility and cost of the strategy. In general, standardizing and training should always be used, and repetition is guaranteed to improve precision when it is feasible and affordable.

▪ ACCURACY

The accuracy of a variable is the degree to which it represents the true value. Accuracy depends on both maximizing precision and minimizing bias (Table 4.3). When serum bilirubin was measured using instruments that had been miscalibrated, for example, the results were precise but biased (7). This concept is further illustrated in Figure 4.2. Accuracy and precision do often go hand in hand and many of the strategies for increasing precision also improve accuracy.

TABLE 4.3 THE PRECISION AND ACCURACY OF MEASUREMENTS

	PRECISION	ACCURACY
Definition	The degree to which a variable has nearly the same value when measured several times	The degree to which a variable approximates the true value
Best way to assess	Comparison among repeated measures	Comparison with a “gold standard”
Value to study	Increase power to detect effects	Increase validity of conclusions
Threatened by	Random error (chance) contributed by	Random and systematic error (bias) contributed by
	The observer	The observer
	The participant	The participant
	The instrument	The instrument

Unlike precision, accuracy is affected by **bias** (systematic error); the greater the error, the less accurate the variable. The three main classes of random error noted in the earlier section on precision each have corresponding biases here.

- **Observer bias** is a distortion, conscious or unconscious, in the perception or reporting of the measurement by the observer. It may represent systematic errors in the way an instrument is operated, such as a tendency to round blood pressure measurements down to a nearest zero mm Hg or to use leading questions when interviewing a participant.
- **Instrument bias** can result from faulty function of a mechanical instrument. A scale that has not been calibrated recently may have drifted downward, producing consistently low body weights.
- **Participant bias** is a distortion of the measurement by the study participant, for example, in reporting an event; it is also called “respondent or recall bias.” Patients with breast cancer who believe that alcohol is a cause of their cancer, for example, may exaggerate the alcohol intake they report.

The accuracy of a measurement is best assessed by comparing it, when possible, to a “**gold standard**”—a reference measurement carried out by a technique that is believed to best represent the true value of the characteristic. What measurement to designate as the gold standard can be a difficult judgment, drawing on previous work in the field and a consensus of experts.

For measurements on a continuous scale, the degree of accuracy can be expressed as the mean difference between the measurement under investigation and the gold standard across study participants. For measurements on a dichotomous scale, accuracy in comparison with a gold standard can be described in terms of sensitivity and specificity (Chapter 13). For

measurements on categorical scales with more than two response options, the percent correct on each can be calculated.

▪ **FIGURE 4.2** The difference between imprecision and bias.

Strategies for Enhancing Accuracy

The major approaches to increasing accuracy include the first four strategies listed earlier for precision, and three additional ones (Table 4.4):

1. Standardizing the measurement methods.
2. Training and certifying the observers.

TABLE 4.4 STRATEGIES FOR REDUCING SYSTEMATIC ERROR, WITH ILLUSTRATIONS FROM A STUDY OF ANTIHYPERTENSIVE TREATMENT

	STRATEGY TO REDUCE SYSTEMATIC ERROR	SOURCE OF SYSTEMATIC ERROR	EXAMPLE OF SYSTEMATIC ERROR	EXAMPLE OF STRATEGY TO PREVENT THE ERROR
1.	Standardizing the measurement methods in an operations manual	Observer	Consistently high diastolic blood pressure (BP) readings due to using the point at which sounds become muffled	Specify the operational definition of diastolic BP as the point at which sounds cease to be heard
		Participant	Consistently high readings due to measuring BP right after walking upstairs to clinic	Specify that participant sit in quiet room for 5 min before measurement
2.	Training and certifying the observer	Observer	Consistently high BP readings due to failure to	Trainer checks accuracy of observer's reading by

			follow procedures specified in operations manual	repeating the measurement
3.	Refining the instrument	Instrument	Consistently high BP with standard cuff in participants with very large arms	Use extra-wide BP cuff in obese or very muscular patients
4.	Automating the instrument	Observer	Conscious or unconscious tendency for observer to read BP lower in group randomized to active drug	Use automatic BP measuring device
		Participant	BP increase due to proximity of attractive technician	Use automatic BP measuring device
5.	Making unobtrusive measurements	Participant	Tendency of participant to overestimate compliance with study drug	Measure study drug level in urine
6.	Calibrating the instrument	Instrument	Consistently high BP readings due to manometer being out of adjustment	Calibrate every month
7.	Blinding	Observer	Conscious or unconscious tendency for	Use double-blind placebo to conceal

		observer to read BP lower in active treatment group	study group assignment
	Participant	Tendency of participant who knew she was on active drug to overreport side effects	Use double-blind placebo to conceal study group assignment

3. Refining the instruments.
4. Automating the instruments.
5. **Making unobtrusive measurements.** It is sometimes possible to design measurements that the participants are not aware of, thereby eliminating the possibility that they will consciously bias the variable. For example, an evaluation of the effect of placing a hand sanitizer and a hand hygiene poster in a hospital cafeteria utilized observers who blended in with cafeteria customers (8).
6. **Calibrating the instrument.** The accuracy of many instruments, especially those that are mechanical or electrical, can be increased by periodic **calibration** with a gold standard.
7. **Blinding.** This classic strategy does not improve the overall accuracy of the measurements, but it can eliminate differential bias that affects one study group more than another. In a double-blind clinical trial, if blinding is successful the participants and observers do not know whether active medicine or placebo has been assigned, and any inaccuracy in measuring the outcome will be the same in the two groups.

The decision on how vigorously to pursue each of these seven strategies for each measurement rests, as noted earlier for precision, on the judgment of the investigator. The considerations are the potential impact that the anticipated degree of inaccuracy will have on the conclusions of the study, and the feasibility and cost of the strategy. The first two strategies (standardizing and training) should always be used, calibration is needed for any instrument that has the potential to change over time, and blinding is essential whenever feasible.

▪ VALIDITY

Validity resembles accuracy, but we think of it as adding a qualitative dimension to considering how well a measurement represents the phenomena of interest. For example, blood levels of creatinine and cystatin C, two chemicals excreted by the kidneys, might be equally *accurate* (eg, within 1% of the true level), but cystatin C may be more *valid* as a measure of kidney function because creatinine levels are also influenced by the amount of muscle (9).

Validity is often not amenable to assessment with a gold standard, particularly for measurements aimed at subjective and abstract phenomena such as pain or quality of life. Several constructs are used to address the validity of these measurement approaches.

- **Content validity** examines how well the measurement represents all aspects of the phenomena under study. For example, the SF-36 questionnaire includes questions on social, physical, and emotional functioning, general health, mental health, and bodily pain to assess quality of life (3).
- **Face validity** describes whether the measurement seems inherently reasonable, such as assessing frailty by muscle weakness, feeling exhausted, weight loss, and slow usual walking speed (10).
- **Construct validity** is the degree to which a certain measurement matches what one would expect from theory; for example, a measurement of social isolation should be able to separate individuals who have few or no friendships and

who do not participate in social activities from those who have many friends and participate in several social activities.

- **Predictive validity** is the ability of the measurement to predict an outcome; for example, how well the measurement of frailty predicts admission to long-term care.
- **Criterion-related validity** is the degree to which a new measurement, such as a new biochemical test of muscle mass, correlates with measurements of muscle strength.

The general approach to measuring subjective and abstract phenomena is to begin by searching the literature and consulting with experts in an effort to find a suitable instrument (typically a questionnaire) that has already been validated. Using such an instrument has the advantage of making the results of the new study comparable to earlier work in the area and may simplify and strengthen the process of applying for grants and publishing the results. Its disadvantages,

however, are that the validation process may have been suboptimal and that an instrument taken off the shelf may be outmoded or not optimal for the research question.

If existing instruments are not suitable for the needs of the study, then the investigator may decide to develop a new measurement approach and validate it herself. This can be an interesting challenge and even lead to a worthwhile contribution to the literature, but it generally requires a lot of time and effort. It is fair to say that the process is often less conclusive than the word “validation” connotes.

▪ OTHER FEATURES OF MEASUREMENT APPROACHES

Measurements should be **sensitive**[2] enough to detect differences in a characteristic that are important to the investigator. Just how much sensitivity is needed depends on the research question. For example, for a trial of whether a treatment improves heart function, measuring the distance that a participant is able to walk in 6 minutes is likely to be more sensitive to change in cardiac performance than a self-reported rating of shortness of breath after walking one block.

An ideal measurement is **specific**, representing only the characteristic of interest. For example, an echocardiographic measurement of the contractile function of the heart would be more specific to the effect of a treatment on heart failure than the 6-minute walking distance, which is also influenced by lung function and the fatigability of leg muscles.

Measurements should be appropriate to the objectives of the study. A study of stress as an antecedent to myocardial infarction, for example, would need to consider what kind of stress (psychological or physical, acute or chronic) was of interest before setting out the operational definitions for measuring it.

Measurements should provide an **adequate distribution of responses** in the study sample. A measure of functional status is most useful if it produces values that range from high in some participants to low in others. A major reason for pretesting is to ensure that the actual responses do not all cluster around one end of the possible range of responses, also described as “floor and ceiling effects.”

Whenever possible, measurements should be designed in a way that minimizes subjective judgments. **Objectivity** is achieved by reducing the involvement of the observer and the participant and by using automated instruments. One danger in these strategies, however, is the consequent tunnel vision that limits the scope of the observations and the ability to discover unanticipated phenomena. This can be addressed by including some **open-ended questions**, and an opportunity for acquiring subjective and qualitative data, in addition to the quantitative measurements.

In designing a study, there is a tendency to keep adding items that are not central to the research question but that could be of interest. It is true that additional measurements enable investigators to use the data to answer additional questions. However, it is important to keep in mind the value of **efficiency** and **parsimony**. The full set of measurements should be designed to collect useful data at an affordable cost in time and money with acceptable burden on participants. Collecting too much information is a common error that can tire participants, overwhelm the team making the measurements, and clutter data management and analysis. The result may be a more expensive study that is paradoxically less successful in answering the main research questions.

▪ MEASUREMENTS ON STORED MATERIALS

Some measurements can be made only during contact with the study participants, but many can be carried out later on biologic specimens banked for chemical, genetic, or other analyses or on images from radiographic and other procedures filed electronically (Table 4.5).

TABLE 4.5 EXAMPLES OF MEASUREMENTS THAT CAN BE MADE ON STORED MATERIALS

TYPE OF MEASUREMENT	EXAMPLES	BANK FOR LATER MEASUREMENT
Medical history	Diagnoses, laboratory test results, medications, operations, symptoms, physical findings; application of natural language processing	Electronic medical records
Biochemical measures	Markers of inflammation, levels of cell-free DNA, microbiome profile	Serum, plasma, urine, stool
Genetic/molecular tests	Whole genome sequencing, DNA methylation	Whole blood or tissue specimens
Imaging	Body composition, bone structure, artificial intelligence analyses to predict disease or outcomes	X-rays, computerized tomography, and other methods

One advantage of such storage is the opportunity to reduce the cost of the study by making measurements only on individuals who turn out during follow-up to have an outcome of interest and a sample of others, such as by doing a **nested case-control study** (Chapter 9). Archiving biologic specimens and images also has the advantage that scientific advances after the study has begun may lead to new ideas and measurement techniques that can then be employed, funded by new grants. Specimens should be collected carefully, frozen quickly, and archived in a way—at very low temperatures (−70 °C or even −190 °C) that preserves them for future assays. This approach allows for the application of increasingly sophisticated types of measurements, such as proteomics or expression of genes in individual cells. Investigators should consider archiving several types of specimens such as plasma, urine, white blood cells, or stool samples for quantifying species of bacteria, to anticipate development of a wide array of future assays. It is also important to obtain informed consent from participants that covers a wide scope of future potential uses of the specimens.

▪ SUMMARY

- 1. Variables are either **numerical** or **categorical**. Numerical variables are **continuous** (quantified on an infinite scale) or **discrete** (quantified on a finite scale such as integers); categorical variables are **nominal** (unordered) or **ordinal** (ordered). Those that have only two categories are termed **dichotomous**.
- 2. Variables that contain more information provide greater power and/or allow smaller sample sizes, according to the following hierarchy: **continuous variables > discrete numerical variables > ordinal variables > nominal** and **dichotomous variables**.
- 3. The **precision** of a measurement (ie, the reproducibility of replicate measures) is another major determinant of power and sample size. Precision is reduced by **random error** (chance)

from three sources of variability: the observer, the participant, and the instrument.

- 4. Strategies for **increasing precision** that should be part of every study are to **define** and **standardize methods** in an **operations manual** and to train and certify observers. Other strategies that are often useful are **refining** and **automating** the instruments and **repetition**—using the mean of repeated measurements.
- 5. The accuracy of a measurement is the degree to which it approximates a gold standard. Accuracy is reduced both by **imprecision** (random error) and **bias** (systematic error) from the same three sources: the observer, participant, and instrument.

- 6. The strategies for minimizing bias include all those listed for precision (except repetition). In addition, bias can be reduced by making **unobtrusive measures** and by **calibrating instruments**. In comparisons between groups, blinding reduces differential bias.
- 7. **Validity** is the degree to which a measurement represents the phenomena it is intended to measure. It can sometimes be assessed by comparison to a **gold standard**. For abstract and subjective variables where there is no gold standard, it is assessed by **content validity, face validity, construct validity, predictive validity,** and **criterion-related validity**.
- 8. Individual measurements should be **sensitive, specific, appropriate,** and **objective**, and they should produce a **range of values**. In the aggregate, they should be broad but parsimonious, serving the research question at moderate cost in time and money and not placing undue burden on participants.
- 9. Investigators should consider **storing biologic samples, images,** and **other materials** for later measurements that can take advantage of new technologies as they are developed and the efficiency of nested case-control designs.

[1]When there are two measurements of a continuous variable per participant, it may be tempting to express their agreement using a correlation coefficient. However, because the correlation coefficient is extremely sensitive to outliers, a better approach is a “Bland-Altman” plot (4), in which the difference between the two measurements is plotted as a function of their mean. If the absolute value of the difference between the measurements tends to increase linearly with the mean, the coefficient of variation is a better way to summarize variability than the within-participant standard deviation.

[2]We’ll define sensitivity and specificity more formally as characteristics of diagnostic tests in Chapter 13. Here, they have the same general meaning: sensitivity relates to how good the measurement is at finding what you want to find, whereas specificity is about how good it is at avoiding finding what you don't want to find.

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Appendix 4A

Operational Definition of a Measurement of Grip Strength

The **operations manual** describes the method for conducting and recording the results of all the measurements made in the study. This example describes the use of a dynamometer to measure grip strength. To standardize instructions from examiner to examiner and from participant to participant, the protocol includes a script of instructions to be read to the participant verbatim.

▪ PROTOCOL FOR MEASURING GRIP STRENGTH WITH THE DYNAMOMETER

Grip strength will be measured in both hands. The handle should be adjusted so that the participant holds the dynamometer comfortably. Place the dynamometer in the right hand with the dial facing the palm. The participant's arm should be flexed 90° at the elbow with the forearm parallel to the floor.

1. Demonstrate the test to the participant. While demonstrating, use the following description: "This device measures your arm and upper body strength. We will measure your grip strength in both arms. I will demonstrate how it is done. Bend your elbow at a 90° angle, with your forearm parallel to the floor. Don't let your arm touch the side of your body. Lower the device and squeeze as hard as you can while I count to three. Once your arm is fully extended, you can loosen your grip."
2. Allow one practice trial for each arm, starting with the right if she is right-handed. On the second trial, record the kilograms of force from the dial to the nearest 0.5 kg.
3. Reset the dial. Repeat the procedure for the other arm.

The arm should not contact the body. The gripping action should be a slow, sustained squeeze rather than an explosive jerk.

Appendix 4B

Exercises for Chapter 4. Planning the Measurements: Precision, Accuracy, and Validity

1. Classify the following variables as dichotomous, nominal, ordinal, continuous, or discrete numerical. Could any of them be modified to increase power, and, if so, how?
   a. History of heart attack (present/absent)
   b. Age

c. Education (college degree or more/less than college degree)

d. Race

e. Number of alcohol drinks per day

f. Depression (none, mild, moderate, severe)

g. Percent occlusion of coronary arteries

h. Hair color

i. Obese (body mass index ≥ 30 kg/m^2)/nonobese (body mass index < 30 kg/m^2)

2. 2. An investigator is interested in the research question: “Does intake of fruit juice at age 6 months predict body weight at age 1 year?” She plans a prospective cohort study, measuring body weight using an infant scale. Several problems (listed further on) are noted during pretesting. Are these problems due to imprecision (random error), bias (systematic error), or both? Is the problem mainly due to observer, participant, or instrument variability, and what can be done about it?

   a. The 10-kg reference weight used to calibrate the scale has a true weight of 10.2 kg.

   b. Weighing the 10-kg reference on the scale repeatedly gives a mean ± standard deviation of 10.01 ± 0.2 kg.

   c. Some babies are frightened, and when they try to squirm off the scale, the observer must hold them on to complete the measurement.

   d. The pointer on the scale swings up and down wildly for babies who don’t lie still.

   e. Some of the babies arrive for the examination immediately after being fed, whereas others are hungry; some of the babies are weighed with wet diapers.

3. 3. The investigator is interested in studying the effect of resident work-hour limitations on surgical residents. One area she wishes to address is burnout, and she plans to assess it with two questions (answered on a 7-point scale) from a more extensive questionnaire: (a) “How often do you feel burned out from your work?” and (b) “How often do you feel you’ve become more callous toward people since you started your residency?”

   The investigator sets out to assess the validity of these questions for measuring burnout. For each of the following descriptions, name the type of validity being assessed:

   a. Residents with higher burnout scores were more likely to drop out of the program in the following year.

   b. A small group of residents are asked to review the items and report that these seem like reasonable questions to ask to address burnout.

   c. Burnout scores increase during the most arduous rotations and decrease during vacations.

   d. A previous study of more than 10,000 medical students, residents, and practicing physicians showed that these two items almost completely captured the emotional exhaustion and depersonalization domains of burnout as measured by the widely accepted (but much longer) Maslach Burnout Inventory (1).

Reference

1. West CP, Dyrbye LN, Sloan JA, Shanafelt TD. Single item measures of emotional exhaustion and depersonalization are useful for assessing burnout in medical professionals. *J Gen Intern Med*. 2009;24(12):1318-1321.

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